TEXTBOOK TITLE

An example textbook template with margin notes

Contents

Chapter 1

The first chapter

1.1 Lorem ipsum

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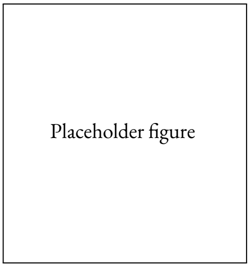

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[1] This is an example footnote.

[1] L. Li Tianjun, W. Xia, W. You-kai and Z. Shou-hua, *Distinguishing the Color Octet Axial-Vector-like Particle for Top Quark Asymmetry via Color Flow Method at the LHC* (June 2013), arXiv:1306.3586

[2] W. S. McCulloch and W. Pitts, *A logical calculus of the ideas immanent in nervous activity*, The Bulletin of Mathematical Biophysics, 5 (4), 115–133 (1943)

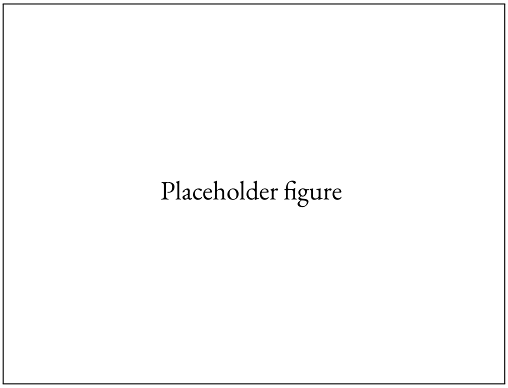

Figure 1.1: Placeholder figure.

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Figure 1.2: HH logo.

1.2 Mathematical symbols $HH \to b\bar{b}\tau^+\tau^-$

The di-Higgs final state $HH \to b\bar{b}\tau^+\tau^-$ features heavy-flavour jets and leptonic or hadronic τ decays.

Equations without references:

$$E = mc^2.$$

Equations with references:

$$\int_0^\infty e^{-x^2}\,\mathrm{d}x = \frac{\sqrt{\pi}}{2}. \tag{1.1}$$

1.3 Examples of unit typesetting

- Speed of light: $c = 2.997\,924\,58 \times 10^8\ \mathrm{m/s}$
- Beam energy: 6.5 TeV per beam
- Tracker pitch: 50 µm to 75 µm
- Temperature variation: ± 0.3 K
- Luminosity: 2.0/fb
- 12.3 kg
- 3.45×10^8

1.4 Examples of unit typesetting using electron-volt units with better kerning

- Usage in normal text: the beam energy is 6.5 TeV per beam.
- Usage in mathematics environment: $E_{\text{beam}} = 6.5\,\mathrm{TeV}$.
- Usage with `siunitx`: The beam energy is 6.5 TeV per beam.
- The PETRA accelerator beams at DESY were of 24 GeV/*c* each.

1.5 Examples of shorthand commands

- Jets are reconstructed using the anti-k_t clustering algorithm.
- The k_t algorithm clusters softer constituents first.
- A Higgs boson can decay to a $b\bar{b}$ pair.
- A b-hadron contains at least one b-quark.
- Secondary vertices can reveal long-lived b-hadrons.
- The leading b-jet has the highest transverse momentum.
- Events with two b-jets are retained for further analysis.
- A b-quark hadronises before it can be observed directly.
- Two b-quarks are produced in this Higgs-boson decay.
- The b-tagging efficiency is measured in collision data.
- Each selected jet must satisfy the b-tag requirement.
- The event contains exactly two b-tags.
- A c-hadron contains at least one c-quark.
- Misidentified c-hadrons form an important background.
- CPT symmetry relates particles to their antiparticles.
- The η–ϕ separation helps distinguish nearby objects.
- Jets, leptons, photons *etc.* may be reconstructed in the detector.
- Charged leptons, *e.g.* electrons, leave tracks in the inner detector.
- The candidate is neutral, *i.e.* it carries no electric charge.
- A 5σ excess is conventionally associated with a discovery.
- The $H \to b\bar{b}$ channel has the largest Higgs-boson branching fraction.
- The H_{T} variable is the scalar sum of transverse momenta.
- The $H_{\mathrm{T}}^{\mathrm{had.}}$ variable includes only the hadronic contribution.
- A large-R jet can contain all the decay products of a boosted particle.
- Large-R jets are useful for reconstructing boosted heavy particles.
- Large $\not{E}_{\mathrm{T}}$ can indicate invisible particles in the final state.

- The selection may be written as $\not\!\!E_{\mathrm{T}} > 200\,\mathrm{GeV}$.
- The M_{T2} endpoint can constrain the mass scale of a process.
- The leading-lepton p_{T} spectrum is compared with the prediction.
- The p-value quantifies the compatibility of the data with a hypothesis.
- Tracks curve in the r–ϕ plane when a magnetic field is present.
- The weak interaction is based on the $\mathrm{SU}(2)_{\mathrm{L}}$ gauge group.
- The S_{T} requirement suppresses low-energy background events.
- A 3σ deviation is evidence, but not normally a discovery.
- Inclusive $t\bar{t}$ production is a major background at the LHC.
- The $t\bar{t} + b\bar{b}$ process contains an additional b-quark pair.
- The $t\bar{t} +$ jets sample includes events with additional jets.
- The $t\bar{t}b\bar{b}$ final state contains two t-quarks and two b-quarks.
- The $t\bar{t}{+}b\bar{b}$ notation highlights b-jets produced with a t-quark pair.
- The $t\bar{t}H$ process directly probes the t-quark Yukawa coupling.
- The $t\bar{t}H\left(b\bar{b}\right)$ channel combines associated production with a b-quark decay.
- The $t\bar{t}\left(H \to b\bar{b}\right)$ notation makes the Higgs-boson decay explicit.
- The t-quark is the heaviest elementary particle yet observed.
- At the LHC, t-quarks are produced both singly and in pairs.
- A $W^{\pm}$-boson mediates charged-current weak interactions.
- Pairs of $W^{\pm}$-bosons provide a sensitive test of electroweak theory.
- The symbol $W^{\pm}$ includes both charged states of the weak boson.
- A Z^{0}-boson mediates neutral-current weak interactions.
- The symbol Z^{0} denotes the neutral weak boson.

Index

Bibliography

[1] L. Li Tianjun, W. Xia, W. You-kai and Z. Shou-hua, *Distinguishing the Color Octet Axial-Vector-like Particle for Top Quark Asymmetry via Color Flow Method at the LHC* (June 2013), arXiv:1306.3586 1

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